

Title: Identification of a new genetic defect responsible for the polymorphism of (S)-mephenytoin metabolism

PMID: 7969038

DOI: N/A

Abstract: A genetic polymorphism in the metabolism of the anticonvulsant drug (S)-mephenytoin has been well documented in humans. There are marked interracial differences in the frequency of the poor metabolizer phenotype, which comprises 2-5% of Caucasian but 18-23% of Asian populations. We have recently reported that the principal genetic defect responsible for the poor metabolizer phenotype is a single-base pair mutation in exon 5 of CYP2C19 (CYP2C19m), which accounts for approximately 75-83% of the defective alleles in both Japanese and Caucasians subjects. In the present study, we have identified a new mutation (CYP2C19m2) in Japanese poor metabolizers, consisting of a guanine to adenine mutation at position 636 of exon 4 of CYP2C19, which creates a premature stop codon. Genotyping of seven Japanese poor metabolizers who were not homozygous for the previously described CYP2C19m defect (now designated CYP2C19m1) indicated that they were either homozygous for the new defect (CYP2C19m2/CYP2C19m2) or heterozygous (CYP2C19m1/CYP2C19m2) for the two defects. CYP2C19m1 accounts for 25 of 34 alleles in Japanese poor metabolizers, whereas CYP2C19m2 accounts for the remaining nine alleles. Hence, CYP2C19m1 and CYP2C19m2 explain 100% of the available Japanese poor metabolizers (34 alleles). In contrast, the CYP2C19m2 defect was not detected in nine Caucasian poor metabolizers (83% of available poor metabolizer alleles were CYP2C19m1), indicating the existence of another, as yet unidentified, mutation. Genetic testing of the families of two Japanese poor metabolizer probands showed that coinheritance of the CYP2C19m1 and CYP2C19m2 alleles was concordant with the autosomal recessive inheritance of the poor metabolizer phenotype.